Telco Customer Churn Analysis

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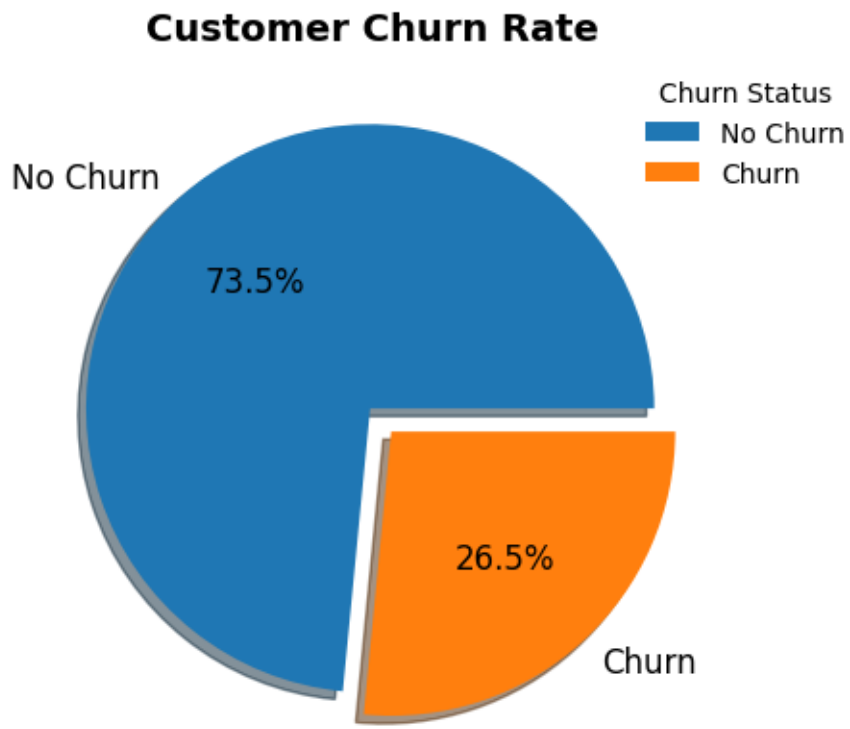
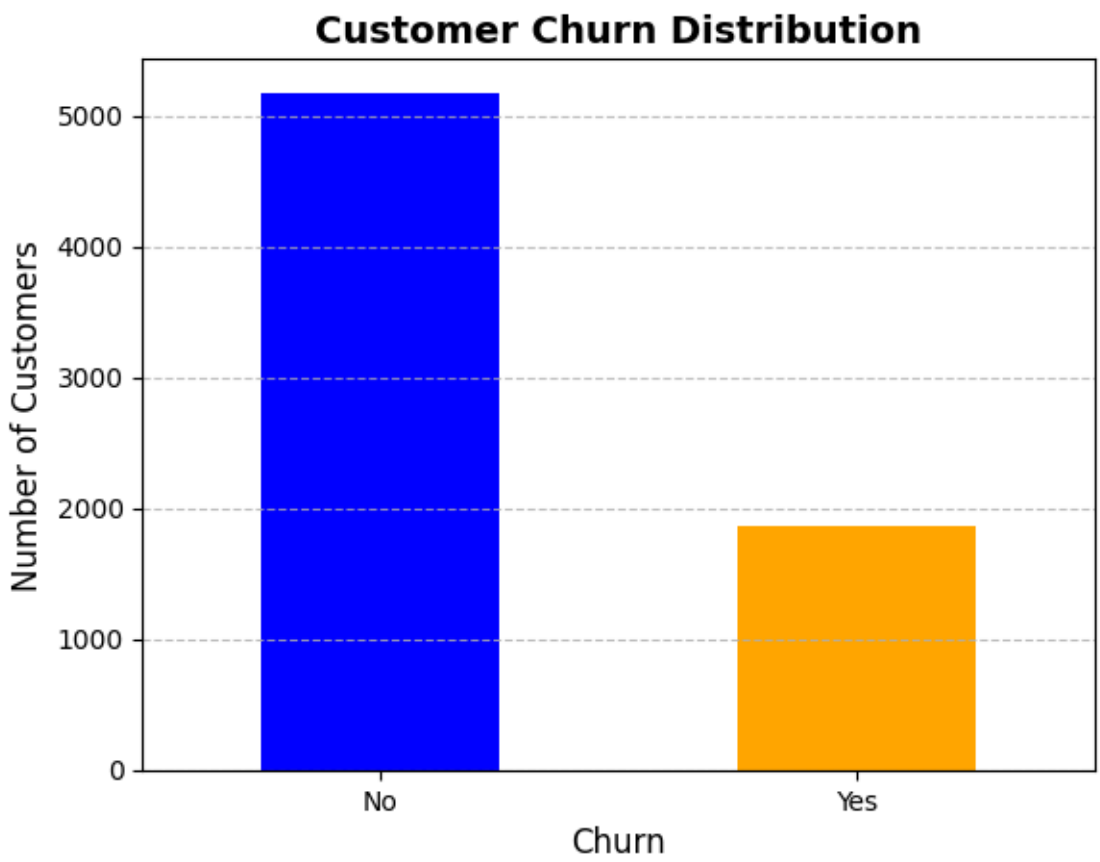
Churn Analysis

Preview

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines
0	7590-VHVEG	Female	No	Yes	No	1	No	No phone servic
1	5575-GNVDE	Male	No	No	No	34	Yes	No
2	3668-QPYBK	Male	No	No	No	2	Yes	No
3	7795-CFOCW	Male	No	No	No	45	No	No phone servic
4	9237-HQITU	Female	No	No	No	2	Yes	No

Overall Churn Rate

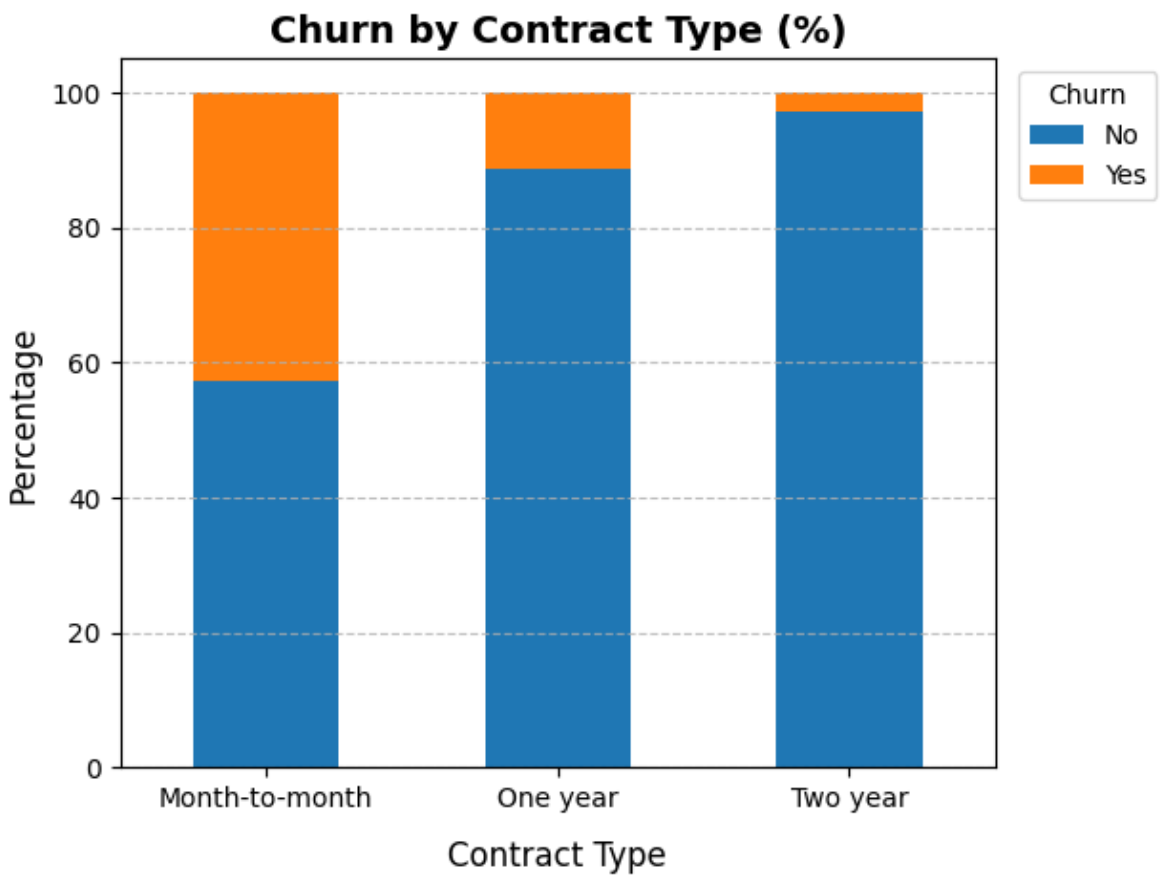




The company is experiencing a relatively high churn rate (1,868 of 7,043 customers) indicating a significant loss of customers and potential revenue. Immediate retention strategies are required, particularly targeting high-risk segments.

Churn by Contract

Contract type is a strong predictor of churn. Customers without long-term commitment are far more likely to leave.



Keys observations: Customers on month-to-month contracts have a significantly higher churn rate, roughly 43%, compared to one-year, which is 11%, and two-year contracts, which is appproximately 3% of all customers in the two year contract.

- **Retention Driver:**
  - Longer commitments equate to lower churn. Customers who sign longer contracts are 15x more likely to stay than month-to-month customers.

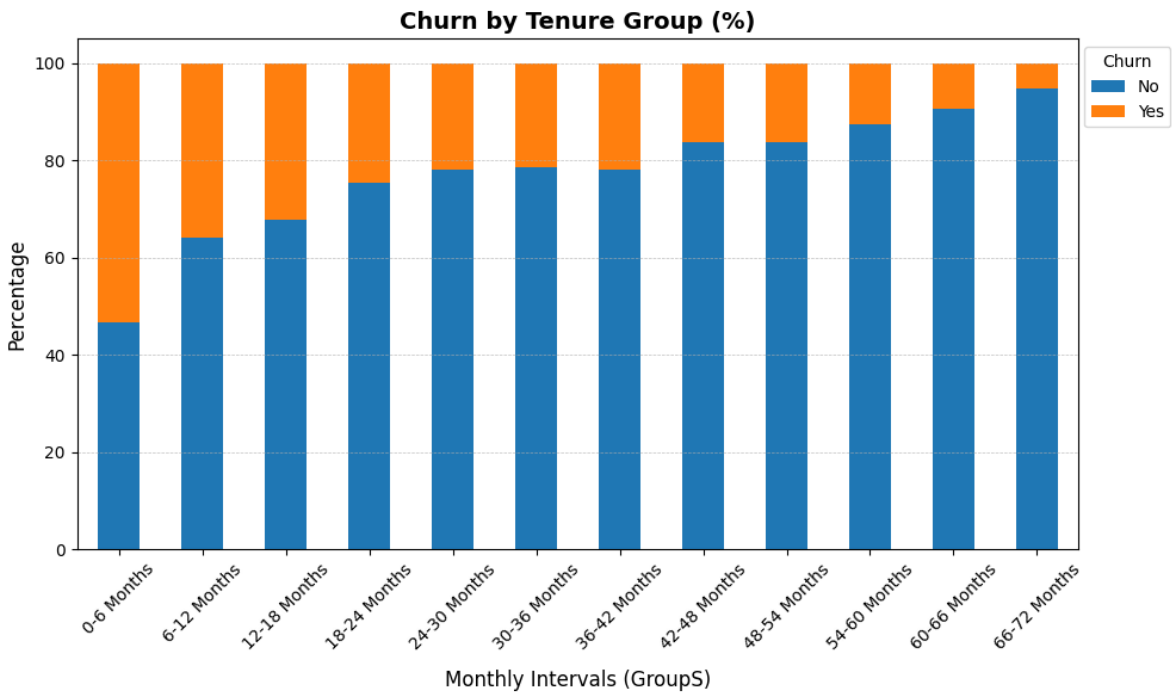
Measure to minimize churn:

- **Targeted Strategies:** Incentives for annual prepayment can be offered, like one month free. This gives the customers something to look forward to.
- Or monthly trials could be utilised to show customer the value the business is to them, (e.g. a "3-month trial" that auto-converts to annual contract)
- Have loyalty rewards

Churn by Tenure

The previous results already show how much the time customers have been with us contrivutes to churn.

The graph below will show the rate of churn and retention with how long a customer has been with the company (in months).



Key observations: New customers are the biggest risk. In the first 6 Months, over half (roughly 53.3%) of the new customers leave. At this point, customers have not committed meaning competitors are still high and the more time it takes the business to be of value to them, the greater the chances of customers leaving increasing churn. Onboarding & early experience matter most.

- **Retention Driver:**
  - Time = loyalty. Once customers pass the 1-year mark, churn drops dramatically

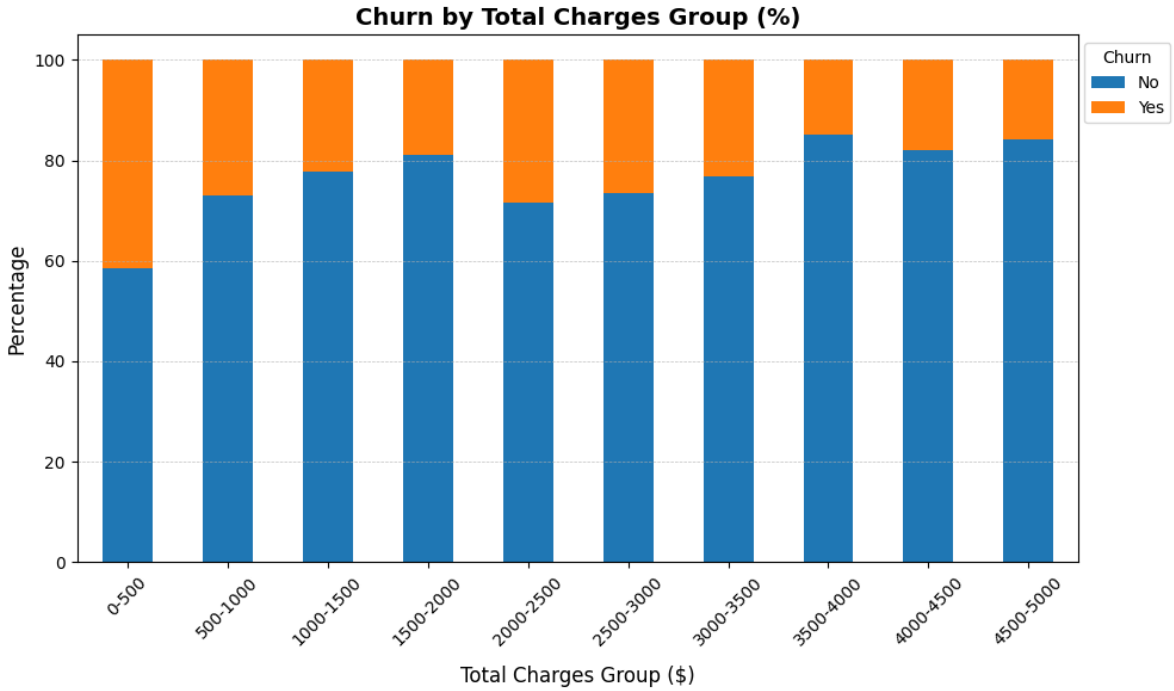
Measures to minimize churn:

- **Targeted Retention Strategies:** Since the first are a risk, identinify the customers not using the service actively and reach out to them with personalized offers or support to encourage continued engagement.

- **Loyalty Programs**: For customers in the mid-tenure groups (24-36 months), reward long-term commitment.
- **Feedback Mechanisms**: For customers in the later tenure groups, establish feedback mechanisms to understand their evolving needs and preferences. This helps retain the loyal customers we have.

Churn by Total Charges

Total charges is a measure of how much revenue a customer has generated for the company. By grouping customers based on their total charges and analyzing the churn rate within each group, we can identify if there are specific spending patterns that correlate with higher or lower churn rates. This can help the company to target retention efforts more effectively, such as offering special promotions or personalized services to high-value customers who may be at risk of churning.



Key Observations: - **Retention Driver:** - Customers who accumulate higher total charges have invested more and tend to stay. - TotalCharges is **NOT** a strong independent driver of churn - **Note on trend:** - The trend is almost horizontal meaning that after initial period, total charges isn't a strong predictor - once customers survive first ~6 months, they're likely to

stay regardless of bill size. - While customers with low total charges exhibit higher churn, this is primarily due to their short tenure. Total charges alone is not a strong predictor of churn, but rather a reflection of customer lifetime

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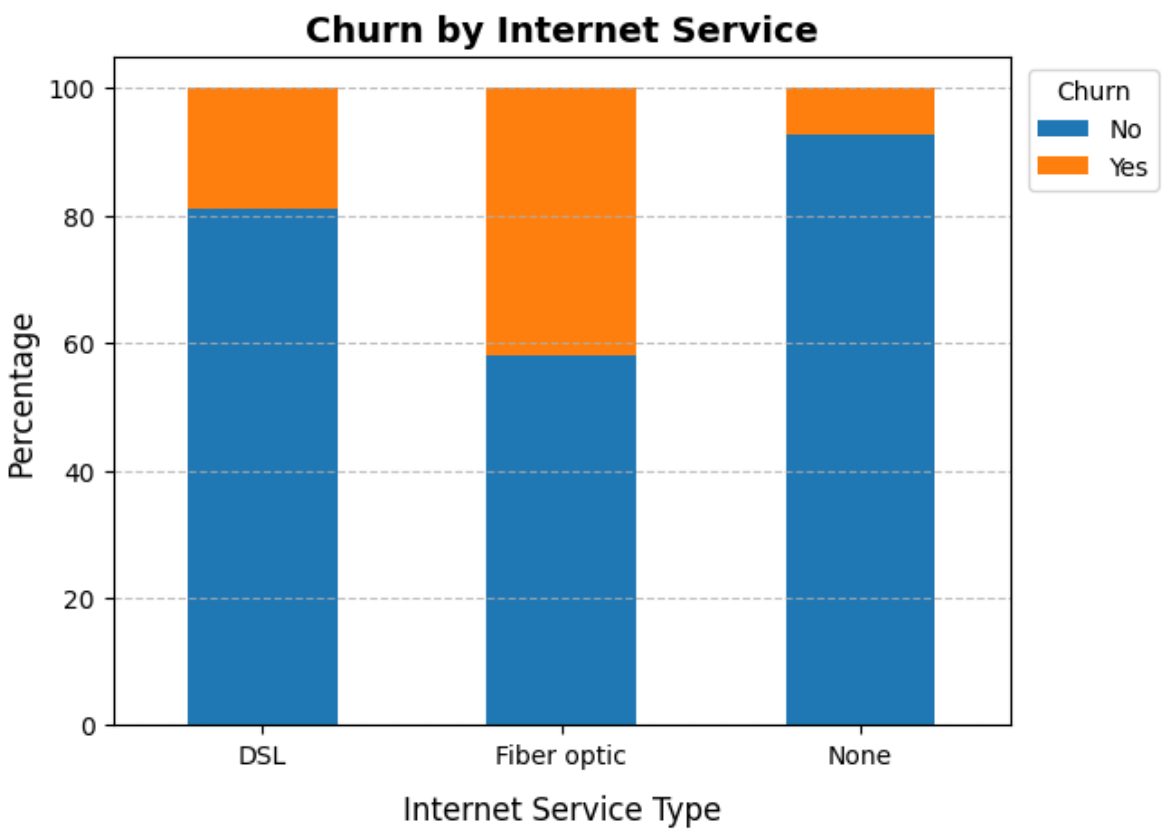
Churn by Internet Service

This section analyzes the churn rate based on the type of internet service customers have. The dataset categorizes internet service into three types: 'DSL', 'Fiber optic', and 'None'. By calculating the percentage of customers who churned within each internet service category, we can identify if there are any significant differences in churn rates among these groups. This analysis can help the company understand if certain internet service types are associated with higher customer dissatisfaction and churn, which can inform targeted retention strategies.

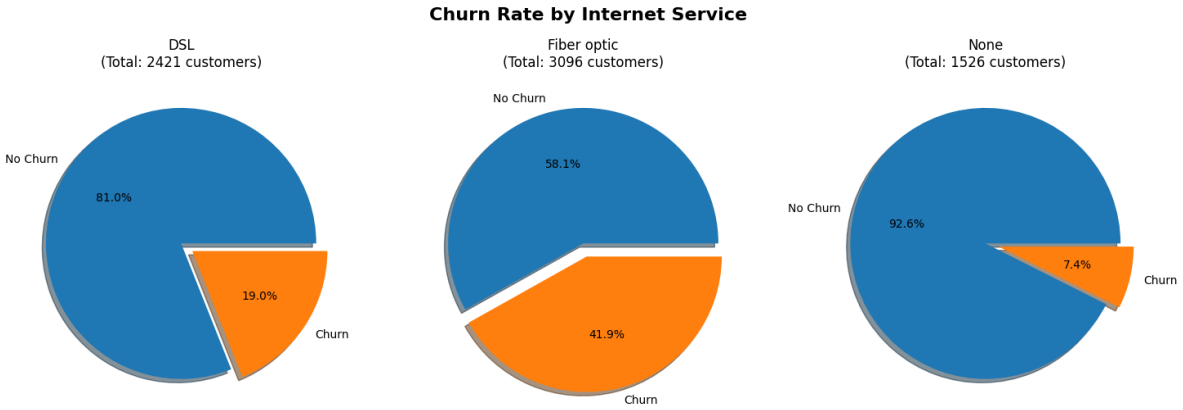


Key Obsertion:

Customers with Fiber optic internet service have a significantly higher churn rate (around 42%) compared to those with DSL (around 14%) and None (around 10%).

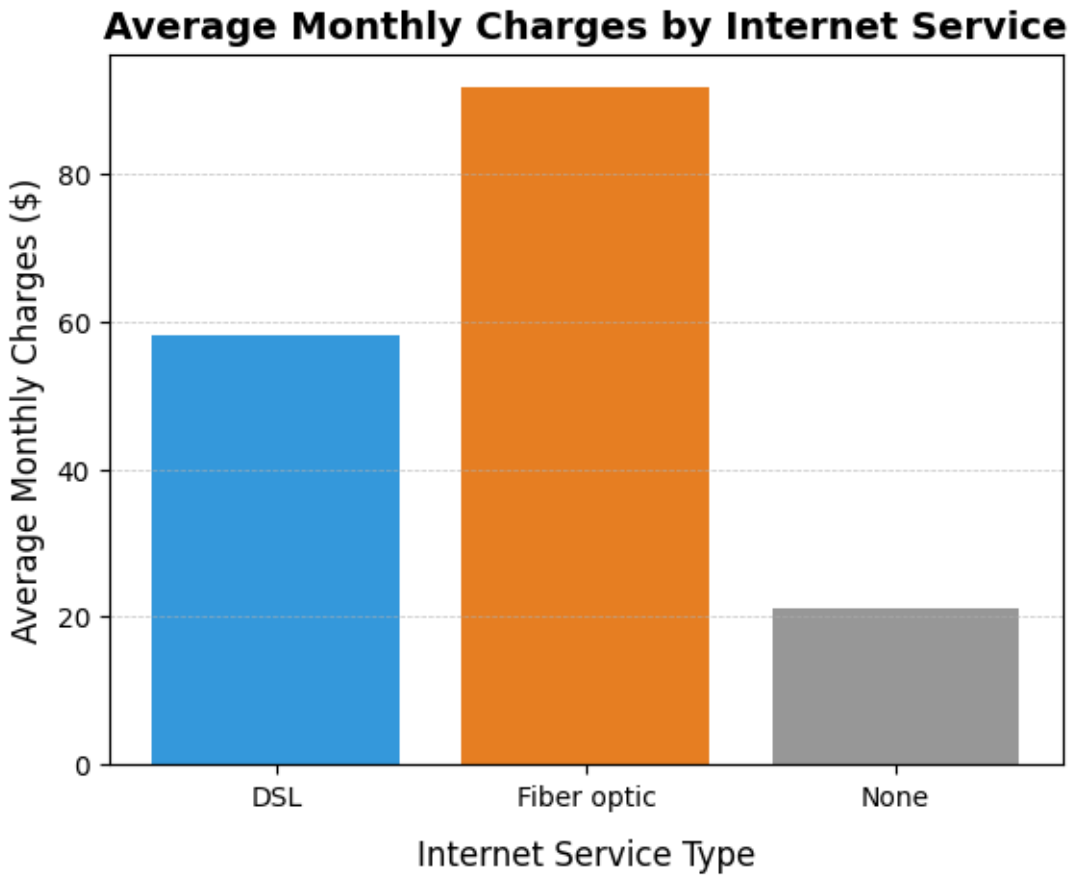
The majority of customers with Fiber optic service are churning, which indicates a potential issue with the service quality or customer satisfaction for that group.

Customers with no internet service (None) have the lowest churn rate, which may suggest that these customers are less likely to churn due to the absence of internet-related issues or dissatisfaction.



Possible Drivers to Investigate: - How customers perceive Fiber Optics - Service reliability - Competitors offering better fiber deals

Recommendation: - URGENT: Survey fiber optic churners to identify what leads to customer dissatisfaction - Survey both DSL and fiber opitcs to compare customers' feedback on both services - Explore potential improvements to both the fiber optic and DSL service, such as enhancing network infrastructure, offering better customer support, or providing incentives to retain customers who are at risk of churning due to dissatisfaction with their internet service. - Guide customers to the RIGHT service for THEIR needs (which service is for what type of user) not just the most expensive or fastest option.



Fiber optic customers pay nearly 60% more than DSL customers, yet they churn at 42% vs DSL's 19%. This suggests that customers pay more, but they're not experiencing enough value to justify the cost.

Summary

The analysis shows that customer churn (**roughly 26.5%**) is primarily **driven by early-stage disengagement, low contract commitment, and fiber optic service usage (expextations not being met)**. Customers in their **first 6 months** and those on **month-to-month plans** are the **most likely to churn**, while **long-term customers** (higher tenure) **demonstrate strong retention**. This highlights a clear customer lifetime trend: the longer a customer stays, the less likely they are to leave. Although total charges appear related to churn, this is **largely due to their correlation with tenure rather than being a direct**

driver. Fiber optic users exhibit the highest churn rates, suggesting potential issues with pricing or service quality. **Overall**, customers tend to stay longer once they pass the early stages, indicating that **improving onboarding, encouraging long-term contracts, and addressing fiber-related concerns** would significantly improve retention.